

Condition Number

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0.1 Eigenvalues and eigenvectors

When we have a square matrix, we can apply it to a vector space to transform vectors, moving them to a different location; most will not remain on the same axis, though some will, these are known as eigenvectors (provided they are non-zero vectors, $u \neq 0$).

Since the eigenvector is not rotated in any way, remaining on its original axis and merely being stretched or contracted, it is equivalent to multiplying that eigenvector by a scalar.

Describing this mathematically:

$$Au = \lambda u$$

Where:

- A is our matrix
- u an eigenvector
- λ an scalar

0.2 Eigendecomposition

If A has n linearly independent eigenvectors, $u_1, u_2, \dots, u_n$ with their respective eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$, for each one the following holds:

$$\begin{aligned} Au_1 &= \lambda_1 u_1 \\ Au_2 &= \lambda_2 u_2 \\ &\vdots \\ Au_n &= \lambda_n u_n \end{aligned}$$

We can pack these vectors into a matrix.

V will be the matrix whose columns are the eigenvectors.

$$V = [u_1 \ u_2 \ \cdots \ u_n]$$

Λ will be the matrix whose main diagonal consists of the eigenvalues:

$$\Lambda = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}$$

When performing matrix multiplication, we can observe...

$$\begin{aligned} AV &= A(u_1 \ u_2 \ \cdots \ u_n) = Au_1 \ Au_2 \ \cdots \ Au_n = \lambda_1 u_1 \ \lambda_2 u_2 \ \cdots \ \lambda_n u_n \\ V\Lambda &= \lambda_1 u_1 \ \lambda_2 u_2 \ \cdots \ \lambda_n u_n \end{aligned}$$

Therefore

$$AV = V\Lambda$$

With this, we can finally solve for A to obtain the general formula for the eigendecomposition.

$$A = V\Lambda V^{-1}$$

0.3 Application to Portfolio Theory

Imagine a three-dimensional vector space where each axis represents an asset, and any point within the space represents a portfolio with weights distributed among these three assets.

Calculating the covariance matrix of the three assets and applying it to the vector space causes the space to rotate and either stretch or shrink, thereby affecting the majority of the vectors in the same way.

However, in this new space, there will be vectors whose direction remains unchanged, undergoing only stretching or shrinking. These correspond to specific portfolios from the original vector space that were diversified in such a way that each is completely independent of the others.

These are our eigenvectors, and their corresponding eigenvalues measure the risk associated with these independent portfolios.

To apply this to our global minimum-variance portfolio, we must invert the covariance matrix.

$$\begin{aligned} \Sigma &= V\Lambda V^\top \\ \Sigma^{-1} &= V\Lambda^{-1}V^\top \end{aligned}$$

Bearing in mind that the diagonal of Λ contains our eigenvalues, upon inversion, its main diagonal becomes:

$$\Sigma^{-1} = \begin{pmatrix} 1/\lambda_1 & 0 & \cdots & 0 \\ 0 & 1/\lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1/\lambda_n \end{pmatrix}$$

This is where the main problem arises, since the direction in the vector space associated with lower risk, and thus a very small eigenvalue, now has a significantly high multiplier within Σ^{-1} .

$$\lambda_1 = 1, \quad \lambda_2 = 0.5, \quad \lambda_3 = 0.001.$$

Inverting

$$1, \quad 2, \quad 1000.$$

0.4 Condition Number

To measure how sensitive the covariance matrix is to this type of change, we can perform a stress test on the matrix and measure its condition number.

$$\kappa(\Sigma) = \frac{\lambda_{\max}}{\lambda_{\min}}$$

If the matrix is well-conditioned and none of the n independent portfolios dominates another, then

$$\lambda_{\max} \approx \lambda_{\min} \therefore \kappa(\Sigma) \approx 1$$

If the matrix is ill-conditioned

$$\lambda_{\min} \rightarrow 0 \therefore \kappa(\Sigma) \rightarrow \infty$$